

2-3 Assignment: Vector Sorting

Code Reflection & Pseudocode

Malgorzata Debska

Professor: David Ostrowski

Code Reflection:

The provided code aims to implement sorting algorithms (specifically selection sort and quicksort) for sorting bids based on their titles extracted from municipal government data. The program reads bid information from a CSV file, allows users to choose between selection sort and quicksort algorithms to sort the bids, and displays the timing results for each sorting operation.

During development, I encountered the challenge of integrating the sorting algorithms into the existing code structure while ensuring proper functionality and adherence to specifications. I approached this by carefully studying the requirements, understanding the logic of selection sort and quicksort algorithms, and incrementally implementing and testing each functionality.

Pseudocode:

READ IN user input and store it in a variable named choice for menu selection.

IF choice equals 1 (**Load Bids**):

- SET a variable named ticks to the current clock time using the clock() method to store the starting clock tick
- INITIALIZE a CSV file
- PARSE the CSV file and store the data into bid objects.
- DEFINE a vector to store a collection of bids.
- RETURN the bids
- PRINT the size of bids
- CALCULATE the elapsed time by subtracting the stored starting clock tick from the current clock time using the clock() method.
- PRINT the elapsed time.
-

ELSE IF choice equals 2 (**Display All Bids**):

- LOOP through the bids vector
- PRINT each bid
- INSERT a break

ELSE IF choice equals 3 (**Selection Sort All Bids**):

- SET a variable named ticks to the current clock time using the clock() method to store the starting clock tick.
- INVOKE the selection sort method and pass in the vector bids.
- PRINT the size of bids
- CALCULATE the elapsed time by subtracting the stored starting clock tick from the current clock time using the clock() method.
- PRINT the elapsed time.
- INSERT a break.

ELSE:

- PRINT "Goodbye".
- RETURN 0.

END PROGRAM.

These pseudocode descriptions capture the essence of the sorting algorithms clearly and understandably. It provides a high-level overview of the steps involved in sorting bids based on their titles. Overall, developing this code provided valuable experience in implementing sorting algorithms and integrating them into existing codebases.